Coordination-Collapse Phase Boundary in Policy-Gradient Cooperative MARL

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Abstract

Cooperative multi-agent reinforcement learning (MARL) is known to collapse into a degenerate *lazy-agent* policy when reward centralization falls too low, but this failure has never been characterized as a quantitative object. We treat it as one: we measure the critical centralization w_{crit} at which coordination gives way to collapse—with a pre-registered collapse definition and bootstrap confidence intervals—turning a qualitative failure mode into a measurable phase boundary, the first in deep cooperative MARL. The boundary is an instrument, and our central finding uses it to characterize how *loss aversion* governs coordination: loss-averse agents ($\lambda > 1$) systematically *raise* w_{crit}—they over-weight the immediate local cost of cooperating against its diffuse, delayed global benefit, so the team needs more reward centralization to coordinate. This is our most strongly supported result: the rightward shift is monotone in λ, saturating, and *replicates across four environment–algorithm settings*: the supply chain under PPO and under centralized-critic MAPPO, MPE Cooperative Navigation under MAPPO, and Level-Based Foraging under MAPPO—i.e. three environments and two algorithms (four of the six possible combinations). On the supply chain a full sweep λ=1...7 is tracked by a one-parameter, theory-bounded law (RMSE 0.014): the shift is $+0.089$ $[0.041, 0.141]$ at λ=3 (within the human range) and up to $+0.166$ $[0.135, 0.202]$ at λ=7 (a high-stakes upper bound). A mean-field signal-to-noise account *fixes this sign a priori* (proved as a sign invariant of the escape barrier) and is independently corroborated by an exploration-noise ablation; both the metastable horizon dependence and the exploration-noise effect further *generalize* beyond the supply chain (metastability to MPE and LBF; the noise effect to LBF, with a mechanism-predicted null on MPE). Two further results sharpen the picture but are scoped to the supply-chain substrate, where they hold cleanly. A three-way decomposition shows probability-weighting curvature moving w_{crit} in the *opposite* direction (left by 0.100 $[0.033, 0.172]$); and—a striking, falsifiable prediction we establish on this one substrate—the boundary is set by a *minority by network position*: a single *downstream* loss-averse node drives $w_{\mathrm{crit}} = 0.853$, indistinguishable from a fully loss-averse population (0.851), whereas the same bias placed upstream moves it far less (0.726). Where a directed resource-flow ordering exists, this predicts a system's centralization requirement is fixed by its most downstream biased participant, not the mean. The boundary doubles as a safety margin: how much more centralization a cooperative system needs once its agents are loss-averse.

Introduction

Cooperative systems increasingly mix human decision-makers with learning agents trained to act like them. In automated replenishment, fraud review, energy dispatch, and clinical triage, the deployed agents do not optimise a clean objective—they inherit human preference structure by design, either because they are trained on human choices through preference-aligned objectives that are themselves prospect-theoretic in form (**?**), or because they act alongside humans in the loop (**?**). The most robustly documented feature of that preference structure is *loss aversion*: people—and now the agents shaped after them—weigh a loss more heavily than an equivalent gain (**??**). Individual-level behavioral economics tells us what this does to one decision-maker. It does not tell us what it does to the *coordination* of many decision-makers learning together (**??**)—which is the question a deployer actually faces.

That coordination has a well-documented way of failing. Cooperative multi-agent reinforcement learning (MARL) collapses into a degenerate, zero-action *lazy-agent* policy when agents abandon cooperation (**??**). Whether a system collapses depends on its *reward centralization* $w \in [0, 1]$—the fraction of each agent's reward drawn from the team's global outcome rather than its own local one: push w too low and each agent's own performance drowns out its contribution to the team, so cooperation carries no learnable signal and the safe, lazy policy wins. Yet the field describes this dependence only qualitatively—"centralization helps" (**??**). *Where*, in w, coordination gives way to collapse—the critical centralization w_{crit} at which the collapse probability crosses one-half—has never been measured. It is a slogan, not a quantity.

We ground the measurement in automated replenishment for multi-echelon supply chains, and the choice is not incidental: it is the kind of domain where the sharpest version of our question can even be *asked*. A supply chain has a *directed flow*—inventory moves downstream along a chain, so "upstream" and "downstream" are well defined—which lets us ask not merely whether a behavioral bias matters but *where in the network* a biased participant matters most. Its failure modes are economically legible (hoarding, stockouts, the bullwhip cascade), so collapse is a recognisable operational disaster rather than an abstract training pathology; and real replenishment is done by planners who demonstrably weigh losses more heavily than gains, so the agents replacing or

assisting them inherit that bias by construction. The substrate thus supplies the directed structure, the legible failure, and the behavioral mechanism in one place—which is why our most fine-grained results live here, and why we are explicit about which of them we claim beyond it.

We treat collapse the way physics treats a phase boundary: a pre-registered, frozen binary collapse definition, logistic estimation of $P(\text{collapse} \mid w)$, and bootstrap confidence intervals on w_{crit}. We use "phase boundary" in the empirical sense—a sharp, metastable threshold whose asymptotic ($T \to \infty$) value is the boundary proper—not as a claim of a thermodynamic-limit singularity. The boundary is then an *instrument*. The decisive design choice that makes what it measures trustworthy is that we apply cumulative prospect theory (CPT) as a *reward transform after reward blending*, never inside the learning update (2): the optimiser is byte-for-byte identical across all conditions, so a measured shift in w_{crit} is a behavioral effect, not an artifact of tuning or of a changed algorithm—separating us cleanly from prior CPT-in-MARL work that folds the value and probability-weighting functions into the actor–critic update (**?**). Against this null, a three-way design (no PT / curvature only / curvature plus loss aversion) attributes any shift in w_{crit} to a specific behavioral component.

What we find is this. *Human-like loss aversion forces cooperative learning agents to require more reward centralization to avoid collapse—a shift we predict in sign a priori and confirm across three environments and two algorithms—and a single downstream loss-averse agent can set that requirement for the whole system.* The shift is already significant at the human-range $\lambda \approx 3$ ($+0.089$ $[0.041, 0.141]$ in required centralization), rising to a high-stakes upper bound of $+0.166$ $[0.135, 0.202]$ at λ=7; its *sign* is fixed before any data by a sign invariant of the policy-gradient escape barrier (Theorem 1); and its *direction* replicates across a supply chain (under PPO and centralized-critic MAPPO), MPE Cooperative Navigation, and Level-Based Foraging. Our contributions form a ladder: (i) the first quantitative, confidence-intervalled phase boundary of coordination collapse in deep cooperative MARL; (ii) the discovery that this boundary is *metastable*, sliding with the training horizon rather than being a fixed constant; (iii) a behavioral decomposition showing the two CPT components move it in *opposite* directions, with the loss-aversion direction generalizing and a *minority by network position* setting the boundary; (iv) an a priori sign guarantee for that direction; and (v) a practical reading of the boundary as a measurable safety margin—how much more centralization a cooperative system needs once its agents are loss-averse, and which participant sets the requirement. Two of these results (the affine λ-law and the downstream-minority cascade) we establish on the supply-chain substrate and scope honestly as such, framing the cascade as a falsifiable, structurally specified prediction for any directed-flow domain.

Preliminaries and Method

In cooperative MARL, N agents share an environment and aim to maximize a joint objective. Centralised Training with Decentralised Execution (CTDE) (**?**) addresses the *credit assignment* problem by allowing a centralised critic during training while agents execute independently. The degree of credit sharing is controlled by a reward mixing weight: a scalar $w \in [0, 1]$ that interpolates between purely local (w=0) and purely global (w=1) reward.

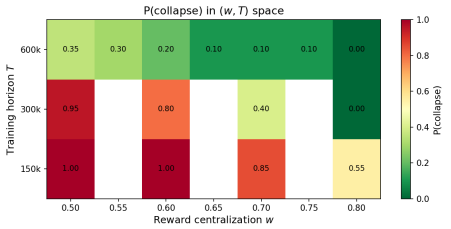


Figure 1: Collapse probability $P(\text{collapse})$ across the (w, T) plane; each cell is the empirical collapse fraction over 20 seeds. As the training horizon T grows, the collapse region recedes to lower w—the *metastable* leftward slide of the boundary. The 150k and 300k rows are risk-neutral (Exp A, α=β=1); the 600k row (Exp T, α=β=0.88) is partially curvature-inflated, and a matched α=β=1 control (Exp 0b) confirms the slide is genuine (§).

When $w \approx 0$, each agent's reward is dominated by its own performance. The contribution of cooperative action is diluted: an agent receives no signal when a teammate's action improves the global outcome. **?** and **?** formalise this as the *lazy agent* problem and *relative overgeneralization* respectively: agents converge to a safe, zero-action policy that avoids coordination risk. This paper asks: at what value of w does this failure mode become probable, and how does agent-level loss aversion shift that value?

Prospect theory (**?**) models human decision-making under uncertainty via a reference-dependent value function and a nonlinear probability weighting function. Cumulative prospect theory (CPT) (**?**) extends this to compound lotteries via rank-dependent integration. The CPT value function is:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0, \end{cases} \tag{1}$$

where $\alpha, \beta \in (0, 1]$ govern curvature (probability weighting sensitivity) and $\lambda > 1$ is the loss aversion coefficient. Setting α=β=1 and λ=1 recovers linear expected utility. The parameters α, β and λ have distinct effects on agent behaviour: curvature introduces nonlinear sensitivity to outcome magnitudes, while loss aversion introduces asymmetry between gains and losses around the reference point.

We apply CPT as a *reward transform* applied after reward blending. The per-step reward for agent i is:

$$r_i^{\text{PT}} = v\big((1 - w)\, r_i^{\text{local}} + w\, r^{\text{global}}\big)\,, \tag{2}$$

where $v(\cdot)$ is given by Equation (1). This is the decisive architectural difference from **?**, the only prior work bringing CPT into cooperative MARL: they fold *both* the value function and the probability-weighting function into the actor–critic update, which entangles the curvature (α, β) and loss-aversion (λ) effects and confounds any behavioral shift with a change to the optimiser. By transforming only the scalar reward

and leaving the policy gradient untouched, our optimiser is *byte-for-byte identical* across all conditions: a measured shift in w_{crit} is therefore a behavioral effect, not an algorithmic one, and the curvature and loss-aversion channels can be dialed independently—which is what makes the component decomposition possible at all.

Measurement method

The reward centralization parameter $w \in [0, 1]$ is the sole structural intervention. All other environment parameters, algorithm hyperparameters, and random seeds are held constant across conditions.

We define coordination collapse as a binary outcome:

$$\text{collapsed}(\text{run}) = \mathbf{1}\left[\bar{S} < 10 \ \wedge \ \bar{\pi} \leq -127.1\right], \quad (3)$$

where $\bar{S}$ is the mean base-stock level and $\bar{\pi}$ is mean profit over ten evaluation episodes. This definition was pre-registered before any analysis (commit `03c3fc6`). Section **??** demonstrates robustness to $\pm 20\%$ threshold perturbation (maximum $\Delta w_{\text{crit}} = 0.015$).

For each experimental condition, we fit:

$$P(\text{collapsed} \mid w) = \sigma(b_0 + b_1 w), \quad (4)$$

where σ is the logistic function. We estimate $w_{\text{crit}} = -b_0/b_1$ (the value of w where collapse probability equals 0.5) and its 95% confidence interval via 2,000 bootstrap resamples at the run level. with Wilson 95% intervals for per-w fractions. For steep transitions (high λ), where the unpenalized MLE slope is ill-conditioned, we use *Firth penalized* logistic regression (Jeffreys prior), stable in this regime; Firth and bootstrap w_{crit} agree throughout.

To attribute a shift in w_{crit} to a specific PT component, we compare three conditions with identical environment and algorithm: **null** (α=β=1, λ=1), **curvature** (α=β=0.88, λ=1: probability weighting, no loss asymmetry), and **loss aversion** (α=β=0.88, $\lambda \in \{3, 7\}$: full PT). The curvature effect is $w_{\text{crit}}(\text{B}, \lambda{=}1) - w_{\text{crit}}(\text{N})$ and the loss-aversion increment $w_{\text{crit}}(\text{B}, \lambda) - w_{\text{crit}}(\text{B}, \lambda{=}1)$; the decomposition is valid because the null removes all PT distortion while curvature adds nonlinearity without loss asymmetry.

Experimental Setup

We use `NetworkManagement-v1` from or-gym (**?**): a six-node multi-echelon inventory network with Poisson demand (μ=20) on the retail edge, 30-period episodes, and uniform lead times. Node 1 (the retail-facing node) follows a fixed demand-pull policy, stabilising the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterised via a base-stock wrapper that maps each agent's continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \text{IP})$ where IP is inventory position. The substrate extends prior DRL inventory work — the Beer Distribution Game Dec-POMDP of **?**, the multi-echelon A3C results of **?**, and single-echelon MARL (**?**) — to a six-node network with per-node agents.

Each agent runs PPO (**?**) with a shared MLP policy (`n_steps=128`, `batch_size=256`) via Stable-Baselines3 (**?**), PettingZoo (**?**), and SuperSuit (**?**) vectorisation. Evaluation is deterministic over 10 episodes with 20 seeds per condition.

Table 1: Experiment summary (same environment and algorithm throughout; α=β=1 is the linear no-PT condition). Total: 1,524 runs.

Exp	Purpose	w grid	λ	α, β	Runs
A@150k	Struct. boundary	0:0.1:1	1	1	220
A@300k	Struct. boundary	0.2:0.8	1	1	140
C	Basin control	0.5, 0.85, 1	1	1	144
N	Null baseline	0:0.2:1	1	1	180
B-Red.	Behav. decomp.	0.5−0.95	1, 3, 7	0.88	300
λ-sweep	Loss-av. law	0.5−0.95	2, 4, 5, 6	0.88	400
T	Metastab. horizon	0.5:0.8	1	0.88	140

Table 1 summarises all seven experiments. All designs were pre-registered before execution. Total run count: 1,524.

Results

Structural Phase Transition and Metastability

Exp A establishes a sharp logistic collapse transition. At 150k steps, $w_{\text{crit}} = 0.822$ $[0.785, 0.858]$ with transition width $\Delta w = 0.278$ (logistic slope $b_1 = -15.78$, $p = 5 \times 10^{-9}$). At 300k steps, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ with $\Delta w = 0.199$ ($b_1 = -22.05$, $p = 3.2 \times 10^{-7}$). In both cases, collapse fractions rise steeply from near-zero to near-one within a $\Delta w < 0.3$ range, consistent with a genuine phase transition rather than a gradual degradation. We use "phase transition" throughout in the empirical sense of a sharp, metastable threshold—an exploration-driven, Kramers-style escape barrier (Section)—and not as a claim of a thermodynamic-limit singularity.

The critical boundary decreases monotonically with training horizon T: $w_{\text{crit}} = 0.822$ at 150k and $w_{\text{crit}} = 0.666$ at 300k (both from Exp A, α=β=1). Figure 1 visualises the collapse surface in (w, T) space. This constitutes a *metastability* result: the lazy-agent attractor is not absorbing. The same leftward slide appears in *both* cross-environment substrates under centralized-critic MAPPO, so metastability is not a supply-chain artifact: on MPE w_{crit} falls $0.82 \to 0.68$ as T grows $2\text{M} \to 8\text{M}$ (slide $+0.14$), and on Level-Based Foraging $0.33 \to 0.20$ as T grows $1\text{M} \to 4\text{M}$ (slide $+0.13$), both with non-overlapping bootstrap intervals (Section **??**). Agents trapped below the isocline at finite T can escape given sufficient training; w controls the energy barrier height of the attractor. The phase boundary proper is the asymptotic ($T \to \infty$) threshold—the deterministic crossing toward which $w_{\text{crit}}(T)$ descends (Equation (10))—so the horizon dependence we observe is the metastable *approach* to that boundary, not evidence against a sharp underlying one. The sharpening transition width (Δw decreasing from 0.278 to 0.199 across 150k to 300k) is consistent with policy convergence reducing outcome variance.

The slide holds under matched α=β=1: a control at 600k (Exp 0b) gives $P(\text{collapse})$ of 0.35/0.20 at w=0.6/0.7 versus 0.80/0.40 at 300k, confirming the horizon effect is real with curvature held fixed (the Figure 1 600k row mixes in the curvature shift; see caption). Initialization is ruled out as a driver: within-w correlations between S_{init} and converged S

are near zero (-0.11 to -0.38; Exp C).

The structural null and its reproducibility

All behavioral shifts are measured against a *single canonical null*: the risk-neutral structural boundary at the 300k decomposition horizon, $w_{\text{crit}}^{\text{null}} = 0.666$ (CI $[0.635, 0.694]$, Exp A @300k), which the dedicated null experiment (Exp N) independently reproduces at 0.685 (difference 0.019, overlapping intervals). Every other null-like number is a labeled offset from this one value, not a separate baseline (**??**): the 0.822 at 150k is the same null at a shorter, metastable horizon, and the 0.586 at $\alpha{=}\beta{=}0.88$ is a *result*—the curvature shift—and the $\lambda{=}1$ anchor of the loss-aversion sweep.

Behavioral Phase Boundary

We sweep loss aversion across the full range $\lambda \in \{1, 2, 3, 4, 5, 6, 7\}$ (all at $\alpha{=}\beta{=}0.88$, 300k steps, 20 seeds per cell), combining Exp B-Reduced at $\lambda \in \{1, 3, 7\}$ with a dedicated fill-in sweep at $\lambda \in \{2, 4, 5, 6\}$. The critical centralization w_{crit} rises with loss aversion across the entire range (**??**, **??**): $w_{\text{crit}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$ for $\lambda = 1, \ldots, 7$. The increase is monotone up to estimation noise at the high-λ plateau: w_{crit} rises strictly from $\lambda{=}1$ to $\lambda{=}5$, then saturates near 0.84–0.85, where the $\lambda{=}5$ and $\lambda{=}6$ estimates (0.839 vs 0.836) are statistically indistinguishable (overlapping CIs).[1] The behavioral claim is unambiguous at the endpoints: $w_{\text{crit}}(\lambda{=}7)$ lower CI 0.845 exceeds $w_{\text{crit}}(\lambda{=}1)$ upper CI 0.624—non-overlapping by 0.221—and the seven-point trend (**??**) shows the relationship is systematic, not a two-point artefact.

Crucially, the high-λ plateau is a *behavioral* saturation, not an artifact of the boundary hitting the $w{=}1$ ceiling: the plateau sits at $w_{\text{crit}} \approx 0.85$ with clear headroom to $w{=}1$, and at $\lambda{=}7$ coordination is *fully* restored well before the ceiling (collapse fraction 0.00 already at $w{=}0.95$). A ceiling artifact would instead pile w_{crit} up against $w{=}1$ with collapse persisting at the highest w; we observe neither. The saturation therefore reflects diminishing marginal sensitivity of the coordination barrier to loss aversion—consistent with the bounded affine law (12), in which $w_{\text{crit}}(\lambda) \to 1$ only as $\lambda \to \infty$—rather than a measurement limit.

The per-w collapse curves for the full sweep appear in App. **??** (**??**).

Table 2 and **??** present the three-way decomposition. Relative to the null ($w_{\text{crit}} = 0.685$):

1. **Curvature effect** ($\alpha{=}\beta{=}0.88$, $\lambda{=}1$): excess $= -0.100$, 95% CI $[-0.172, -0.033]$. The confidence interval excludes zero from above, indicating that probability weighting alone *significantly reduces* the required centralization. Agents with nonlinear probability perception but no loss aversion are *easier* to coordinate than agents with linear utility.
2. **Loss aversion at** $\lambda{=}3$: excess $= +0.089$, 95% CI $[+0.041, +0.141]$ (CI excludes zero; significant).
3. **Loss aversion at** $\lambda{=}7$: excess $= +0.166$, 95% CI $[+0.135, +0.202]$ (CI excludes zero; significant).

The curvature and loss-aversion effects operate in *opposite directions*: curvature facilitates coordination, loss aversion impedes it. At $\lambda{=}7$, the loss aversion increment above the curvature baseline is $+0.265$ ($= 0.851 - 0.586$), overwhelming the curvature benefit of -0.100 for a net shift of $+0.166$ above the structural null (rounding: $-0.100 + 0.265 \approx 0.165 \approx 0.166$).

A Mean-Field Mechanism for the Collapse Boundary

This section explains *why* the boundary exists and how it moves. We claim it at the *sign-and-form level only*: w_{crit} marks a signal-to-noise crossing of the policy gradient, the two prospect-theory components move it in *opposite* directions, and the model predicts these directions, the monotonic ordering, the transition sharpness, and the sign of the horizon dependence *without fitting*—but not the numerical value of w_{crit}, which is calibrated (**??**). The account is the learning-dynamics analogue of the EGT result of **?**: the driver is the gradient signal-to-noise ratio under exploration, not selection over fixed strategies, which makes it specific to gradient-trained neural policies.

Gradient flow and its noise

Each learning agent i runs PPO on a Gaussian policy over its (normalised) base-stock action $a_i \in [0, 1]$ with mean μ_i and exploration scale σ. The shaped reward is the centralisation blend passed through the prospect-theory value function v,

$$r_i = v\big((1-w)\,\ell_i(a_i, a_{-i}) + w\,G(a)\big), \qquad G(a) = \textstyle\sum_j \ell_j, \tag{5}$$

where ℓ_i is node i's local profit and G the system profit. For a Gaussian policy the PPO update on the mean is a stochastic estimate of the return gradient, so the policy mean follows a mean-field Langevin (Fokker–Planck) equation

$$\mathrm{d}\mu_i = \underbrace{\eta\,\partial_{\mu_i}\mathbb{E}\big[(1{-}w)\ell_i + wG\big]}_{\text{drift } D(w)}\,\mathrm{d}t \;+\; \sqrt{2\,\Theta(w)}\,\mathrm{d}W. \tag{6}$$

Because each partner's local profit is itself blended into the global term, the partners' exploration (σ) enters the gradient estimate of agent i through the global channel, so the diffusion carries a centralisation- weighted, cross-agent piece,

$$\Theta(w) \;\propto\; \tfrac{\sigma^2}{B}\big[(1{-}w)^2 s_\ell^2 \;+\; w^2(N{-}1)\,\kappa^2\big], \tag{7}$$

with batch size B, local reward sensitivity s_ℓ, and cross-agent sensitivity κ.

[1] At high loss aversion the transition is very steep and falls between two adjacent grid points—e.g. collapse fractions $1.00, 1.00, 1.00, 0.55, 0.00$ at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$ for $\lambda{=}7$—so the boundary remains identifiable (the $w{=}0.85$ cell is interior, $11/20$ collapsed) but the *unpenalized* MLE slope diverges. We therefore fit every λ with Firth penalized logistic regression (**??**), which is stable under steep transitions; all seven slopes are significant ($p < 10^{-3}$) and the Firth w_{crit} matches the bootstrap estimate to within 0.019. A finer grid ($w \in \{0.80, 0.90\}$) would tighten the high-λ CIs further.

Table 2: Three-way behavioral decomposition of w_{crit} relative to the null model. "Significant" denotes that the conservative difference confidence interval excludes zero. †For λ=7, w_{crit} is from bootstrap only; see footnote in Section .

Condition	w_{crit}	95% CI	Excess	Excess CI	Sig.?
Null (α=β=1, λ=1)	0.685	[0.657, 0.710]	0	—	—
α=β=0.88, λ=1	0.586	[0.539, 0.624]	-0.100	$[-0.172, -0.033]$	✓ (neg.)
α=β=0.88, λ=3	0.774	[0.751, 0.798]	$+0.089$	$[+0.041, +0.141]$	✓
α=β=0.88, λ=7†	0.851	[0.845, 0.858]	$+0.166$	$[+0.135, +0.202]$	✓

The boundary as a risk-dominance crossing

We instrumented the substrate directly (finite-difference of ℓ_i and G along the base-stock axis), and the asymmetry it reveals gives the drift a concrete game-theoretic reading. The continuous action induces a stochastic coordination game whose payoff topology shifts with w. At low w the *lazy* policy is the dominant strategy: an upstream node's inventory is a *local cost* (holding and procurement reduce its own profit). As w grows, a cooperative equilibrium emerges because that same inventory is a *global benefit* (it averts downstream stockouts), turning the environment into an (analogue of a) Stag Hunt. But the cooperative equilibrium emerging does not mean agents will *learn* it: if agent i explores cooperatively (raising a_i) while its partners stay lazy, it pays the local cost yet collects no global flow benefit—a continuous analogue of the *sucker's payoff*. Writing the linearised marginal incentives as $\partial_a \mathbb{E}\ell \to -c$ (the expected sucker penalty) and $\partial_a \mathbb{E}G \to +g$ ($c, g > 0$), the drift toward cooperation is

$$D(w) = \eta\big[\, w\, g - (1-w)\, c \,\big]. \tag{8}$$

What governs whether the optimiser actually climbs toward cooperation is not the drift alone but its ratio to the gradient noise: escape from the lazy basin requires the cooperative equilibrium to become *effectively risk-dominant*—its basin of attraction wide enough for exploration noise to bridge—

$$\mathrm{SNR}(w) \;=\; \frac{D(w)}{\sqrt{\Theta(w)}} \;\gtrsim\; \kappa_c, \tag{9}$$

for an $\mathcal{O}(1)$ critical value κ_c set by the training budget. The cooperative signal grows only with centralisation ($\propto w$), so for small w the right-hand region of (8) is swamped by the exploration variance (7) and the lazy policy is the effective attractor (Figure 2a). The crossing $\mathrm{SNR}(w) = \kappa_c$ defines w_{crit}. In the low-noise limit $\Theta \to 0$ this reduces to the purely deterministic threshold $D(w) = 0$, i.e.

$$w^0_{\text{crit}} = \frac{c}{c+g}, \tag{10}$$

the asymptotic ($T \to \infty$) low-noise *shadow* of the boundary; finite exploration noise raises the operative boundary above it. The drift zero $D(w)$=0 is not directly observable—a preregistered probe finds the conditioned drift weakly positive with no clean zero-crossing, an artifact of basin geometry (Section **??**)—so we read the boundary as a *noise-bridged* risk-dominance crossing, not a drift-sign flip. The coefficients c, g, σ, κ_c are not derived from the cost matrix; the model is calibrated, and its content is the *functional* predictions below.

Sharpness and metastability

Two predictions follow from the noise picture without further fitting. First, escape is a Kramers first-passage problem with mean time $\tau \sim \tau_0 \exp(\Delta U/\Theta)$, exponentially sensitive to SNR; the collapse probability therefore switches abruptly as w crosses w_{crit}—an abrupt, metastable threshold, matching the steep logistic fits of Section . Second, collapse at finite horizon means $\tau > T$; as T grows, a smaller SNR (hence smaller w) suffices to escape, so $w_{\text{crit}}(T)$ decreases monotonically toward (10). This is exactly the observed metastable slide $w_{\text{crit}} = 0.822 \to 0.666$ from 150k to 300k steps, following the $\propto -\ln T$ Kramers trend (Figure 2c).

Formal account: what is predicted versus calibrated

Let $x \in [0,1]$ be a macroscopic *order parameter*—the fraction of agents in the coordinating configuration—evolving under the policy-gradient flow as the Itô SDE

$$\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, N)}\,\mathrm{d}W_t, \tag{11}$$

where F is the mean-field drift of (8) (so $\partial_w F > 0$ and $\partial_\lambda F < 0$ follow from the payoff structure, not assumption: w weights the cooperation gradient $+g$ while λ scales only the loss domain) and $\Sigma(x, N) > 0$ is the gradient-noise variance. Here x is *latent*—not recoverable from policy statistics (Section **??**)—so the claims below concern its law, not a measured trajectory. Three results delimit what the mechanism predicts a priori. Two of them (**??**, mean-field criticality as $N \to \infty$; **??**, the finite-(N,T) calibration gap) establish that the *magnitude* of w_{crit} is calibrated, not predicted; we state and prove them in App. **??**. The third—the one the central claim rests on—we give here.

Theorem 1 (Sign invariance of the loss-aversion shift)**.** *Assume $\Sigma(x, N) > 0$ and $\partial_\lambda F(x, w, \lambda) < 0$ for all $x \in [x^*_{\text{saddle}}, x^*_{\text{coord}}]$. Then the empirical boundary shifts strictly in the direction the mean-field drift predicts: $\partial_\lambda w^{\text{Emp}}_{\text{crit}} > 0$.*

Proof. Write (**??**) as a forward integral, $\Delta U = \int_{x^*_{\text{saddle}}}^{x^*_{\text{coord}}} 2F(z, w, \lambda)/\Sigma(z, N)\,\mathrm{d}z$ (the two minus signs in (**??**)—the integrand sign and the reversed limits—cancel). Differentiating under the integral, the fixed-point locations contribute nothing to first order ($F = 0$ there), so

$$\partial_\lambda \Delta U = \int_{x^*_{\text{saddle}}}^{x^*_{\text{coord}}} \frac{2\,\partial_\lambda F(z, w, \lambda)}{\Sigma(z, N)}\,\mathrm{d}z < 0,$$

the inequality holding *pointwise* because $\Sigma > 0$ and $\partial_\lambda F < 0$ on the whole interval—independently of the sign of F itself.

A lower barrier shortens the escape time, $\partial_\lambda \tau < 0$; symmetrically $\partial_w F > 0$ raises the barrier, $\partial_w \tau > 0$. Applying the implicit function theorem to $G(w, \lambda) = \tau(w, \lambda) - T = 0$,

$$\frac{\mathrm{d}\, w_{\mathrm{crit}}^{\mathrm{Emp}}}{\mathrm{d}\lambda} = -\frac{\partial_\lambda \tau}{\partial_w \tau} = -\frac{(-)}{(+)} > 0.$$

Thus the *magnitude* depends on Σ (**??**), but the *direction* is guaranteed by the mean-field drift alone—the precise sense in which the mechanism is predictive. The identical argument with ∂F signs reversed for concave–convex curvature gives the opposite shift, yielding the predicted directional divergence of the two prospect-theory components. □

Opposite signs of the two prospect-theory components

The mechanism's most distinctive prediction is qualitative and requires no fitting: probability weighting and loss aversion shift w_{crit} in *opposite* directions, and the risk-dominance reading makes the asymmetry concrete. Concave–convex curvature ($\alpha, \beta < 1$) compresses the magnitude of both gains and losses, smoothing the extremes of the stochastic sucker's payoff; this lowers the outcome variance that enters Θ, *raises* the SNR, and hence *lowers* w_{crit}. Loss aversion ($\lambda > 1$) instead magnifies the downside of the sucker's payoff, deepening the perceived lazy basin, inflating Θ and lowering the SNR, which *raises* w_{crit}. The signs are fixed by the value function before any data are seen, and they match the central empirical finding (Section): curvature alone shifts w_{crit} left by 0.100, loss aversion shifts it right by up to 0.166.

Derived form of the loss-aversion law

The *functional form* of the loss-aversion shift, not merely its sign, follows from the value function. Differentiating $\mathbb{E}[v(r)]$ and splitting the marginal mass into gain and loss domains, the loss domain is the only part scaled by λ, so the effective local term is affine in λ:

$$w_{\mathrm{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \qquad \phi(\lambda) = 1 + \rho\,(\lambda - 1), \quad (12)$$

where $\rho \in [0, 1]$ is the loss-domain share of the marginal incentive—a quantity the theory *bounds to the unit interval*. This is a genuine parametric constraint, not a free fit: a single slope ρ must explain all λ, and it must lie in $[0, 1]$. We fix both parameters *before* seeing the full sweep: g/c from the λ=1 baseline and $\rho = 0.61$ from the three Exp B-Reduced points ($\lambda \in \{1, 3, 7\}$), both of which satisfy the bound. Held fixed and tested *out of sample* against the remaining four points ($\lambda \in \{2, 4, 5, 6\}$; the full seven-point sweep in **??**), this single-slope affine law gives

$$w_{\mathrm{crit}}^{\mathrm{pred}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\},$$
$$w_{\mathrm{crit}}^{\mathrm{meas}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\},$$

an RMSE of 0.014 across all seven points (**??**)—a one-parameter, theory-bounded law tracking the entire $w_{\mathrm{crit}}(\lambda)$ curve, not merely the three calibration points (re-fitting ρ to all seven gives $\rho = 0.63$, RMSE 0.013). The lone residual is a mild overshoot at the λ=6, 7 plateau: the leading-order law

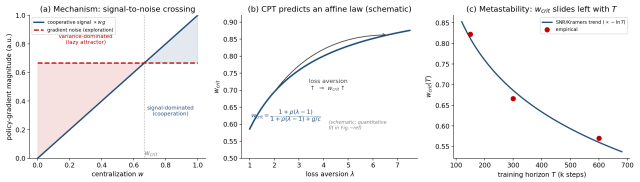


Figure 2: Mean-field mechanism versus empirics. (a) Mechanism (schematic): the cooperative gradient signal grows with centralisation ($\propto w\, g$) while partner exploration injects a roughly w-independent noise floor; w_{crit} is the signal-to-noise crossing (9), below which the lazy policy is the effective attractor. (b) Schematic of the *derived affine* loss-aversion law (12): CPT predicts $\phi(\lambda) = 1 + \rho(\lambda - 1)$ with slope $\rho \in [0, 1]$; the single data-implied slope $\rho = 0.61$ respects the bound. The quantitative fit against the full seven-point sweep is shown separately in **??**. (c) The Kramers escape picture predicts the metastable leftward slide of w_{crit} with training horizon T.

is affine while the data saturate mildly, a second-order effect we do not model. We report this derived affine law rather than a better-fitting but unmotivated power law because the point is mechanism, not goodness of fit.

Scope. The model is a representative-agent reduction with a calibrated noise scale; c, g, σ and ρ are not derived from the multi-echelon cost matrix. A direct finite-difference probe of the substrate confirms the *sign* of the local-cost/global-benefit asymmetry but shows the marginal coefficients are operating-point dependent, so the absolute boundary is a property of the stochastic multi-agent dynamics rather than a static gradient—which is why we calibrate rather than predict its magnitude. This section is not the paper's contribution— the decomposition is—but a supporting, falsifiable *mechanism*: it commits in advance to the signs, the ordering, the affine λ-law with a bounded slope, the sharpness, and the horizon and exploration-noise directions (**??**), all of which the data bear out, consistent with exploration-driven accounts of cooperation failure in shared-policy MARL (**?**). The exploration-noise prediction is confirmed directly by the entropy ablation (Section **??**). Section **??** reports a direct test that measures the Langevin drift and diffusion from PPO training: it reproduces the matched-horizon boundary but leaves the magnitude calibrated, as the drift-sign observable is confounded by basin geometry. We therefore claim this mechanism at the *sign-and-form level only*—the opposite signs, the monotone ordering, the affine form, and the horizon and exploration-noise directions—and treat the numerical value of w_{crit} as calibrated throughout. No result in this paper rests on a predicted magnitude; upgrading the account from calibrated to predictive is left to future work.

Generalization across Environments and Algorithms

The mechanism of Section commits, before any data, to a *direction*: loss aversion inflates gradient noise and therefore

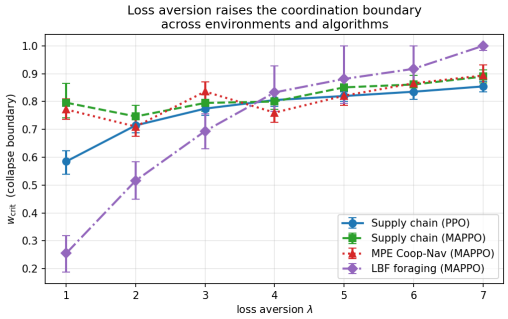


Figure 3: **The loss-aversion effect generalizes.** Critical centralization w_{crit} versus loss aversion λ for four settings: the supply chain under PPO (blue) and under centralized-critic MAPPO (green), MPE Cooperative Navigation under MAPPO (red), and Level-Based Foraging under MAPPO (purple). Error bars are bootstrap 95% CIs. Across three structurally distinct environments and two algorithms, w_{crit} rises with λ then saturates—loss aversion raises the centralization a cooperative system needs to avert collapse. The *direction* is what generalizes; absolute levels differ (most at λ=1) and the curvature sign and exact functional form are substrate-specific. (LBF's high-λ points saturate against the centralization ceiling and are reported as exploratory; see Section .)

raises w_{crit}. If that account is right rather than an artefact of one substrate or one optimiser, the sign should survive changing both. We test this on two further environments and a second algorithm. The headline result is that the *direction* of the loss-aversion effect is robust to both, while the absolute boundary, the curvature sign, and the precise functional form are substrate- and algorithm-specific—exactly the split the calibrated-not-predicted mechanism (**??**) anticipates. **??** (Appendix **??**) records the breadth of evidence behind every headline claim. Figure 3 summarises the headline: in all four settings w_{crit} climbs with λ, the curves converging in the mid-range and differing most at λ=1.

Across environments: incentive misalignment across domains

We test whether the boundary—and the loss-aversion direction—is a recurring feature of incentive-misaligned multi-agent systems rather than a supply-chain artefact, in two structurally distinct environments. Both share the reward blend $r_{\text{blend}}(i) = (1-w)\, r_{\text{local}}(i) + w\, r_{\text{global}}$ but reach collapse by different routes: the supply chain by *resource competition* (hoarding starves downstream nodes) and MPE Cooperative Navigation (`simple_spread_v3`) by *coordination failure* (three agents herd together and cover no landmark). For MPE, collapse is measured by per-step landmark occupancy—a behavioural metric invariant to reward scale, hence unaffected by the CPT transform. Because MPE's coordination reward is *shared-global*, independent learners confound behavioural collapse with a credit-assignment failure; we therefore train it with centralized-critic MAPPO (**?**) (actor sees local observation plus a one-hot agent ID, critic sees the joint state—textbook CTDE), matching the algorithm to the environment's learnability so the measured boundary reflects the incentive w, not the learner's capacity to assign credit. We set the herding scale so the null boundary lands interior (w_{crit}=0.74) and run the full sweep $\lambda \in \{1,\dots,7\}$ at α=β=0.88 (1,760 runs; Figure 3, per-w curves in App. **??**).

The *loss-aversion* effect replicates. On MPE (full λ=1…7 sweep, 20 seeds per cell), w_{crit} rises with loss aversion from 0.78 (λ=1) to 0.89 (λ=7)—rising in envelope but non-monotone at intermediate λ (the λ=2, 4 points dip to within seed noise of the null), so the effect is read as a direction, not a strict ordering. The loss-aversion excess over the null is $+0.158$ $[+0.108, +0.212]$ at λ=7 and $+0.101$ $[+0.055, +0.150]$ at λ=3, both significant and the latter quantitatively matching the supply-chain $+0.089$. The qualitative law—loss aversion raises the centralization required to avert collapse—thus holds across structurally distinct environments and training algorithms (a third environment, Level-Based Foraging, is added in Section ; Figure 3).

The *curvature* effect does *not* replicate. On MPE the curvature excess is $+0.042$ $[-0.009, +0.092]$, statistically indistinguishable from the null and opposite in sign to the supply-chain -0.100; probability weighting appears to facilitate coordination only in the supply-chain substrate. The exact *functional form* of the loss-aversion law is likewise environment-specific: the affine $w_{\text{crit}}(\lambda)$ that tracks the supply chain to RMSE 0.014 fits MPE only to RMSE 0.05, the MPE curve being steeper at high λ as the boundary approaches the centralization ceiling. We therefore claim cross-environment generality for the collapse boundary and for the *direction* of the loss-aversion effect—the paper's central behavioural claim—while reporting the curvature sign and the precise functional form as supply-chain-specific.

A third environment: Level-Based Foraging

MPE shares MAPPO's shared-global reward; to probe a third, structurally distinct domain we add Level-Based Foraging (LBF), a standard cooperative-MARL benchmark (**?**) with a *public-goods* incentive: agents pay an effort cost to load food cooperatively, so the locally rational action is to free-ride and the team coordinates only as the global reward comes to dominate. LBF is natively discrete and unrelated to logistics or navigation; we train it under centralized-critic MAPPO—matching MPE, so that across MPE and LBF the *algorithm* is held fixed and only the *environment* varies—with the identical reward pipeline and full λ-sweep, and a collapse threshold set by the same collapsed-vs-coordinated midpoint used for MPE (1,600 runs, $w \in \{0,\dots,0.95\}$ including the w=0.90 resolver).

LBF yields the most complete decomposition outside the supply chain (Figure 3). The risk-neutral null has a low boundary ($w_{\text{crit}} \approx 0.08$); curvature moves it *right* by $+0.18$—*opposite* to the supply chain's -0.10, again showing the curvature sign is substrate-specific. Loss aversion then drives a large rightward shift: w_{crit} rises from 0.26 (λ=1) to 0.83 (λ=4), monotone with non-overlapping low- versus high-λ intervals (the pre-registered criterion). For $\lambda \geq 5$ the boundary saturates against the centralization ceiling and we report

those points as exploratory (a w=0.90 resolver sharpens the mid-λ estimates but does not change this). As in MPE, the *direction* replicates while curvature sign and functional form do not: the affine law misfits LBF by RMSE 0.15 (versus 0.014 on the supply chain), because the LBF boundary rises more steeply in λ. The *method*—a CI'd boundary that loss aversion shifts—generalizes; the affine *constants and form* are substrate-specific.

Across algorithms: MAPPO on the supply chain

The MPE result varies the environment *and* the algorithm at once. To isolate the algorithm, we re-run the full loss-aversion sweep on the *identical* supply-chain substrate, grid, seeds, collapse label, and hyperparameters, changing only the learner from independent PPO to centralized-critic MAPPO (**?**) (700 runs; $\lambda \in \{1, \ldots, 7\}$, α=β=0.88). The loss-aversion law replicates: w_{crit} rises with loss aversion from 0.796 [0.742, 0.860] at λ=1 to 0.889 [0.856, 0.917] at λ=7, monotone from λ=2 on, with non-overlapping λ=1 vs λ=7 intervals (Figure 3; per-w curves in App. **??**). As in MPE, the *direction* generalizes while the absolute level does not—the MAPPO boundary sits above PPO at low λ (where its transition is shallow and noisy) and the two converge by $\lambda \geq 4$ ($|\Delta w_{\text{crit}}| \leq 0.04$)—so we report the direction and monotonicity as algorithm-independent, and the absolute w_{crit} as algorithm-dependent.

The mechanism's noise prediction: a dissociation

The same mechanism makes a second, sharper cross-environment prediction. Two of its commitments—that the boundary slides left with the training horizon (metastability) and right with injected exploration noise—should themselves generalize, but the noise effect *only where injected entropy is the binding term* in the signal-to-noise ratio (9). Both bear out, and the second as a dissociation. Metastability replicates: the horizon slide appears on MPE (w_{crit} 0.82 → 0.68) and LBF (0.33 → 0.20), both with non-overlapping intervals. The exploration-noise effect replicates on LBF (w_{crit} rising 0.13 → 0.37 as the entropy coefficient grows) but is a *mechanism-predicted null* on MPE: across a sweep extended past the seed-noise floor the boundary does not move (pooled slope β=+0.47, 95% CI [−3.4, +4.3]; 1,320 runs). This is not a failed replication. MPE's categorical actions and dense, distance-shaped reward carry high *intrinsic* exploration that swamps the added entropy, so (9) predicts insensitivity there; the supply chain and LBF have sparse, sharp landscapes where injected noise binds. The effect appears exactly where the mechanism says it should and is absent exactly where it says it should be—a stronger test than uniform replication (numbers in App. **??**).

Two ablations confirm the boundary is contingent on the ingredients the mechanism names: it disappears under coarse discretisation of the continuous control (removing the policy gradient) and is measurable only with a credit-assignment-capable learner on shared-reward tasks (App. **??**). Finally, the boundary is not an averaged property: a single *downstream* loss-averse node can set the centralization requirement of the whole network, a cascade that propagates upstream along the ordering chain (App. **??**; discussed below).

Discussion

That probability weighting and loss aversion exert opposite effects on the coordination boundary is, to our knowledge, novel; prior work applies prospect theory as a unified model without decomposing its components (**??**). Our null-model design isolates them: on the supply chain the beneficial curvature effect (−0.100, substrate-specific) is masked when combined with loss aversion, whose centralization-raising effect dominates and is the component we establish as general. The operational reading is concrete: a system with λ=7 agents—bracketing human loss-aversion estimates (**?**)—requires $w > 0.851$ for reliable coordination against $w > 0.685$ for risk-neutral agents, so the additional 0.166 (Table 2) is a quantified cost of deploying loss-averse agents.

Two caveats sharpen this reading. On calibration, the canonical laboratory estimate of loss aversion is $\lambda \approx 2.25$ (**?**); our sweep deliberately brackets the empirical human range, with λ=7 an upper, high-stakes bound rather than a typical value. The cost scales with λ (Table 2), so a domain whose participants sit nearer $\lambda \approx$ 2–3 incurs a smaller but still positive margin (the λ=3 excess is +0.089, against +0.166 at λ=7). On deployment, centralization is not free: privacy, communication cost, and organisational policy can cap the attainable w. Where the required w exceeds that cap, the boundary reframes the design problem—instead of only raising centralization, one can lower the agents' effective loss aversion (for instance through the training objective), treating λ as the second, behavioural lever the decomposition exposes.

The boundary is also not an averaged property. App. **??** shows it is set by a *minority*: a single downstream loss-averse node moves the system boundary as far as a fully loss-averse population, cascading upstream along the ordering chain. The operational consequence is sharp—in a mixed human–agent supply chain, the required centralization is governed by the most downstream loss-averse participant, not by the population mean.

The sliding w_{crit} is a positive finding: the lazy attractor is *metastable*, so agents below the isocline at finite T can coordinate with more training—consistent with **?**—and the minimum w therefore depends on the training budget.

Our framing sits within a recognised line of multiagent phase-transition results but differs in object: **?** chart a phase diagram for IQL over *environmental* axes, and **?** prove a metastable boundary in opinion dynamics under a *fixed update rule*; ours is the deep-MARL learning-dynamics analogue, a boundary set by the policy-gradient signal-to-noise ratio that a behavioral bias then moves.

Several limitations bound these claims. *Substrate scope:* the full quantitative characterization (decomposition, affine λ-law, metastability, cascade) uses one substrate, chosen because its continuous, per-agent economic reward makes w_{crit} finely identifiable; we replicate the boundary and the loss-aversion *direction* in two further environments (MPE, LBF) and under a second algorithm on the same substrate, but the curvature sign and the exact functional form do *not* transfer, so we claim only the loss-aversion direction as cross-substrate. *Sweep scope:* the steep high-λ transitions are

resolved by a single interior grid point (handled with Firth fits, **??**); the metastability result is finite-horizon; and the heterogeneity sweep uses binary $\lambda \in \{1, 7\}$. *Method scope:* the approach needs an environment learnable without experience sharing and a collapse label separable from task-unlearnability, excluding sparse, long-credit-horizon tasks. *Cascade scope:* the downstream cascade is single-substrate by structural necessity—it is well-posed only where a directed resource-flow ordering, genuine local-vs-global tension, and a learning-reachable boundary co-occur, a rare conjunction most benchmarks violate (App. **??**); we therefore frame it as a falsifiable, structurally specified prediction rather than an established generality.

Conclusion

We presented the first quantitative phase diagram of cooperative MARL coordination collapse, characterising the critical centralization threshold w_{crit} as a function of reward structure, training horizon, and prospect-theoretic loss aversion. Our central and most strongly supported finding is that *loss aversion* impedes coordination—raising w_{crit} ($+0.089$ at λ=3, within the human range; up to $+0.166$ at λ=7)—a direction we predict mechanistically a priori and confirm across three environments and two algorithms. Scoped to the supply-chain substrate, two further results sharpen the picture: a three-way decomposition shows probability-weighting curvature acting in the *opposite* direction (easing coordination by 0.100), and the boundary is set by a downstream loss-averse *minority* rather than the mean bias. We present the latter as a striking single-substrate result and frame it as a falsifiable prediction for any directed-flow domain, rather than a claim shown to generalize. These results provide concrete design constraints for deploying cooperative MARL in behavioral settings and establish a pre-registered phase-diagram methodology applicable to any cooperative MARL system.

Natural extensions include replication across further algorithms (QMIX) and environments (SMAC), upgrading the mean-field account (Section) from a calibrated mechanism to a quantitative prediction of w_{crit} via direct measurement of the policy-gradient signal-to-noise ratio (Section **??**), adaptive centralization scheduling based on the estimated isocline, and characterization of heterogeneous-λ populations.